Adelle Jane Goodwin

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EDUCATION

Monash University, Melbourne, Australia
Doctor of Philosophy (Astronomy & Astrophysics) 2017 –
Thesis: On the Nature of Neutron Stars in Accreting Systems
Supervisors: Prof Duncan Galloway and Prof Alexander Heger

Monash University, Melbourne, Australia
Bachelor of Science Advanced with Honours 2013 – 2017
Awarded first class honours for the thesis: Thermonuclear X-ray bursts from accretion powered millisecond pulsars

AWARDS & PRIZES

Australian Postgraduate Award academic merit based, $81,000, *Monash University* 2017 – 2020

JL Williams PhD top up scholarship academic merit based, $15,000, *Monash University* 2017 – 2020

ICRAR Summer Studentship academic merit based, $6,000
International Centre for Radio Astronomy, Western Australia 2015 – 2016

JL Williams Honours Scholarship academic merit based, $5,000
Monash University 2015

Monash University Extension Studies Program academic merit based,
Monash University 2012

PUBLICATIONS

In 2.5 years I have published 6 peer-reviewed publications including 3 as first-author and 1 as second-author, as well as 3 Astronomer's Telegrams. These publications have accumulated more than 20 citations.

Goodwin A. J., Galloway D. K., Heger A., Cumming A., Johnston Z., *A Bayesian Approach to Matching Thermonuclear X-ray Burst Observations with Models*, 2019, MNRAS, 2266, `https://arxiv.org/pdf/1907.00996`

Goodwin A. J., Galloway D. K., in't Zand J. J. M., Kuulkers E., Bilous A., Keek L., *XMMU J181227.8-181234: a new ultracompact X-ray binary candidate*, 2019, MNRAS, 486, 4149, `https://arxiv.org/pdf/1904.10970`

Goodwin A. J., Heger A., Galloway D. K., *Neutrino Losses in Type I Thermonuclear X-Ray Bursts: An Improved Nuclear Energy Generation Approximation*, 2019, ApJ, 870, 64, `https://arxiv.org/pdf/1808.02225`

Goodwin A. J. and Woods T. E., *The binary evolution of SAX J1808.4–3658: Implications of an evolved donor on the progenitor system*, 2019, MNRAS (in preparation, pre-print avail. upon request)

Galloway D. K., Johnston Z., **Goodwin A.**, Heger A., 2019, IAUS, 121, IAUS..339

Janowiecki S., Cortese L., Catinella B., **Goodwin A. J.**, 2018, MNRAS, 476, 1390

Galloway D. K., **Goodwin A. J.**, Keek L., 2017, PASA, 34, e019

Astronomer's Telegrams:
Goodwin A. J., Russell D. M., Galloway D. K., in't Zand J. J. M., Heinke C., Lewis F., Baglio M. C., 2019, ATel, 12993, 1

Russell D. M., **Goodwin A. J.**, Galloway D. K., in't Zand J. J. M., Lewis F., Baglio M. C., 2019, ATel, 12964, 1

in't Zand J. J. M., Galloway D. K., Kuulkers E., **Goodwin A.**, 2017, ATel, 10567, 1

OBSERVING PROPOSALS

I have applied for observing time with 5 international X-ray space telescopes, with 2 proposals as the principle investigator and 3 as a co-investigator. I have a 100% success rate for these proposals.

INTEGRAL/XMM-Newton, *Co-I, ToO proposal, successful* 2018, 2019

NICER, *PI, ToO proposal, 2 of 3 sources successful* 2019

Swift, *PI, ToO monitoring proposal, successful* 2019

AstroSAT, *Co-I, ToO proposal, successful* 2019

INVITED TALKS

Seminar Speaker, *McGill Space Institute, McGill University* (invited by Prof Andrew Cummings) 2019

I have also been invited to attend two focussed conferences where participation is limited to 20–40 of the world's experts in thermonuclear X-ray bursts. These have been held at the Lorentz Centre (Leiden, Netherlands) and the Monash Prato Centre (Prato, Italy).

CONFERENCE TALKS

ASA annual science meeting, contributed talk, *Sydney, Australia* 2016

International Nuclear Physics Conference, contributed talk, *Adelaide, Australia* 2016

Burst Environment and Nuclear Reactions Meeting, contributed talk, *Prato, Italy* 2018

ANITA conference, contributed talk, *Melbourne, Australia* 2019

Leiden Thermonuclear Burst workshop, contributed talk, *Leiden, Netherlands* 2019

I have also given poster presentations at 3 international conferences.

SERVICE AND OUTREACH

Webmaster
Monash Centre for Astrophysics 2018 – 2020

Conference Organiser
BERN18, Prato, Italy 2018

Conference Organiser
ANITA workshop and summer school, Melbourne, Australia 2019

Conference Organiser
JINA-CEE junior and main conferences, East Lansing, America 2019

School Science Night Volunteer
Frankston Primary School 2017 – 2018

Relay for Life Telescope Volunteer
Murrumbeena, Melbourne, raising money for cancer research 2018

Astrolight festival volunteer
Scienceworks Museum, Melbourne 2017 – 2018

Scientific Expert Panel
Emerging Sciences Victoria, John Monash Science School 2019

TRAVEL GRANTS

Registration fee waiver and per diem confrence funding (merit based)

International Nuclear Physics Conference (Australia)	2016
Nucleii in the Cosmos conference (Italy)	2018
Burst Environment and Nuclear Reactions Meeting (Italy)	2018
Frontiers First Summer School and Frontiers (America)	2019
Lorentz Centre thermonuclear burst meeting (Netherlands)	2019

Astronomical Society of Australia Travel Grant (merit based), *Italy* 2018

JINA-CEE funded travel to visit the NSCL/FRIB at MSU and Prof Hendrick Schatz for 2.5 weeks, *East Lansing, America* 2019

McGill University funded travel to visit Prof Andrew Cumming for 2.5 weeks, *Montreal, Canada* 2019

University of Amsterdam funded travel to visit Prof Anna Watts for 3 weeks, *Amsterdam, Netherlands* 2019

SKILLS

Programming languages
Proficient in Python, Fortran 90, C++, and html/CSS programming. Basic usage of Matlab and Mathematica.

Software experience
Experience using stellar evolution programs such as MESA, Kepler, and Window to the Stars. Experience using X-ray data analysis software HEAsoft.

Telescope Data
I have analysed data from X-ray telescopes including *Swift*, *RXTE*, *INTEGRAL*, *BeppoSAX*, and *XMM-Newton*.

Supercomputer Experience
I have experience using supercomputers such as the Monash Monarch cluster.

TEACHING

Teaching Associate, *Astrophysics and Physics*
I have taught 1st, 2nd, and 3rd year undergraduate astrophysics courses, 1st year undergraduate physics courses, and 1st year undergraduate biophysics courses at Monash University. 2016–2019

Individual Tutor, *Astrophysics*
Indigenous Engagement Unit, Monash University 2018 – 2019

REFERENCES

Ass Prof Duncan Galloway
PhD supervisor
duncan.galloway@monash.edu

Prof Alexander Heger
PhD supervisor
alexander.heger@monash.edu

Prof Andrew Cumming
Collaborator
andrew.cumming@mcgill.ca